Current Language Models Might Not Be Suitable For Reverse Engineering Moral Wisdom of Crowds

Rafal Rzepka, Yuki Katsumata and Kenji Araki

During the first symposium on machine ethics we have proposed an automatic moral judgement algorithm based on measuring polarities of consequences by simple corpus search [1][2]. Although comparatively high precision, the method has two main flaws. First is the limitation of input as only short phrase as “stealing a car” allow retrieving a substantial number of examples for sentiment analysis of what are emotional or social consequences of an act. The second problem was a lexicon-based sentiment analysis tool used for detecting polarities of consequences (words as “wonderful” or “nasty” for emotional realm and “praised” or “arrested” for the social realm), which narrows the scope of consequence analysis to sentences containing the lexicon phrases. The advent of deep learning has allowed to alleviate both difficulties, current methods allow for fuzzy input and matching thanks to contextualized embeddings. To investigate this flexibility, we have randomly chosen 150 simple acts as “a child throws a teddy bear” and asked three crowdworkers to extend these phrases to have a positive (safe) and negative (dangerous) meaning without changing a verb, an actor and patient or object. For example for the teddy bear example a negative version was “a child throws a heavy teddy bear at a baby with force” and positive was “a child throws a light teddy bear on a bed cover” (data extension and the experiments were performed in Japanese language). The experimental results have shown that a BERT [3] language model-based sentiment analysis method tends to recognize positive polarity in most cases: only 5.33% of positively rewritten acts were classified as negative, while 86.2% of negative ones were wrongly recognized as positive. This suggests that also language model-based methods might strongly associate positivity and negativity with words often used in sentiment lexicons. As our data describes acts rarely using clearly subjective words, sentiment analysis module has struggled to differentiate between sentences as “a child throws a teddy bear at the fragile glass thing standing on a high furniture” (positivity 0.94) and “a child throws a teddy bear down on a soft blanket” (positivity 0.95). These results suggest that we should not trust current language models while analyzing positive and negative consequences of acts especially when they represent happenings without using emotive phrases.

[1] Rzepka, Rafal, and Kenji Araki. "What Statistics Could Do for Ethics?: The Idea of Common Sense Processing Based Safety Valve." In AAAI Fall Symposium on Machine Ethics, Technical Report FS-05-06, pp. 85-87. AAAI, 2005.
[2] Rzepka, Rafal, and Kenji Araki. "What people say? Web-based casuistry for artificial morality experiments." In International Conference on Artificial General Intelligence, pp. 178-187. Springer, Cham, 2017.
[3] Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. "BERT: Pre-training of deep bidirectional transformers for language understanding." arXiv preprint arXiv:1810.04805, 2018.